LIG: Layer-wise Integrated Gradients for Within-Layer Flow Analysis in Transformers

MON-19

Poster (PDF) My page

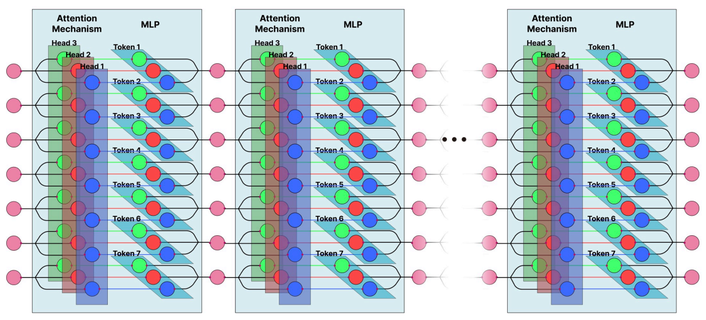
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Attributing ATT and MLP separately and composing them matches whole-layer IG best when their baselines follow the whole-layer baseline.

LIG (Layer-wise Integrated Gradients) applies Integrated Gradients (IG) at the Multi-Head Attention (ATT) and multilayer perceptron (MLP) boundaries and composes them. BERT-base, 1,700 Penn Treebank (PTB) sentences. Best: whole-layer ITB (starts at the target) → ITB for ATT, ATT output at that start for MLP (ATTITBa=0); whole-layer Zero → Zero for both. Agreement, not faithfulness.

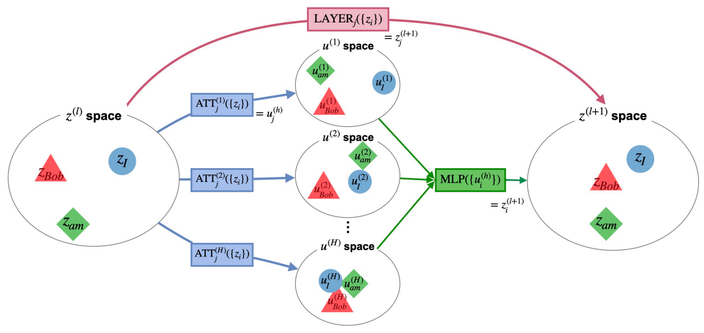
Project page
https://eightsuzuki.github.io/layer-wise-integrated-gradients/

1 What flows inside a layer?



We view each layer as a **dynamic graph**: token representations z_i (pink) and per-head attention outputs $u_i^{(h)}$ (dots on head planes) are nodes; information flows across the ATT and MLP module boundaries.

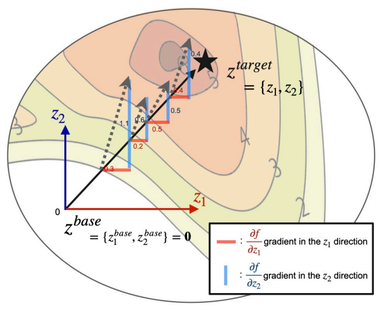
We split each layer into ATT and MLP (bottom path), attribute each boundary with IG, and ask whether their composition matches IG on the whole layer (top path). Shapes: tokens of "I am Bob"; ovals: spaces $z^{(l)}$, $u^{(h)}$, $z^{(l+1)}$.



Method	Attributes	Limitation / LIG's gain
PRIOR Attention map [7]	weights $\alpha_{i,j}$	ignores values and MLP
PRIOR Rollout [5]	attention across layers	ignores values and MLP
PRIOR Norm-based [6]	weighted value norms	no ATT / MLP split
PRIOR AttnLRP [4]	relevance via ATT, MLP	a rule per operation
OURS **LIG**	**IG at each ATT / MLP edge**	**one rule (IG); splits and composes**

Set-to-set IG: use IG to measure how much each element of a module's *input set* (tokens at ATT, heads at MLP) contributes to each output token. **LIG** = **L**ayer-wise **I**ntegrated **G**radients: do this at every ATT and MLP boundary, layer by layer, and compose the results in the spirit of **L**ayer-wise Relevance Propagation (LRP) [2, 3].

2 IG, and where the path starts



PRIOR Integrated Gradients (IG) [1], an explainable-AI (XAI) attribution method, *line-integrates* the gradient along a path from a **baseline** $z(0)$ to the input z; any path gives attributions summing to the output change, so **path** and **start** are ours to choose. Like [1], we use the straight path $z(a) = z(0) + a\,(z - z(0))$, $a \in [0, 1]$, for low cost and simplicity:

$$\mathrm{IG}_d = (z_d - z_d(0)) \int_0^1 \frac{\partial f(z(a))}{\partial z_d}\, da$$

f: scalar output (contours), d: input dimension (figure: z_1, z_2; bars: $\partial f / \partial z_d$ along the path). Completeness: $\sum_d \mathrm{IG}_d = f(z) - f(z(0))$. The baseline sets **what each contribution is measured against**; it can change which ATT / MLP pair agrees best (§5).

3 LIG: set-to-set IG at each boundary OURS

At each boundary (cells below), apply §2's IG to the input x, interpolated from its **baseline** $x(0)$, with $f = \|y_j(a) - y_j(0)\|_2$ (output y_j's L_2 change), then sum over the dimensions of each input element r (token or head); scores are signed:

$$\mathrm{IG}_{r,j} = \sum_{d \in r} \mathrm{IG}_d, \qquad \widetilde{\mathrm{IG}}_{r,j} = \frac{\mathrm{IG}_{r,j}}{\|y_j(1) - y_j(0)\|_2} \quad \left(\sum_r \widetilde{\mathrm{IG}}_{r,j} = 1\right)$$

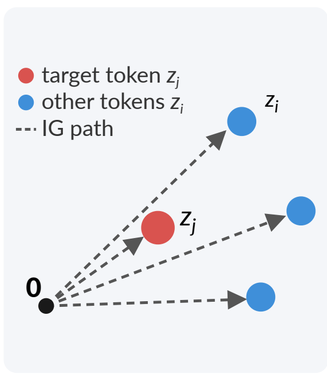
ATT
$z_i \to u_j^{(h)}$: $\mathrm{IG}^{\mathrm{ATT}}_{i,j,h}$

MLP (W_o, LayerNorm, FFN; residual z fixed)
$u_j^{(h)} \to z_j^{(l+1)}$: $\mathrm{IG}^{\mathrm{MLP}}_{h,j}$

Whole layer l
$z_i \to z_j^{(l+1)}$: $\mathrm{IG}^{\mathrm{LAYER}}_{i,j}$

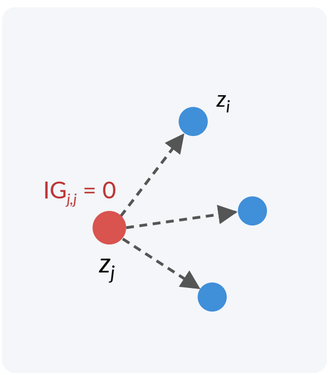
Baselines: how the **input** moves along the IG path, $a = 0 \to 1$ (input-space sketch).

Zero



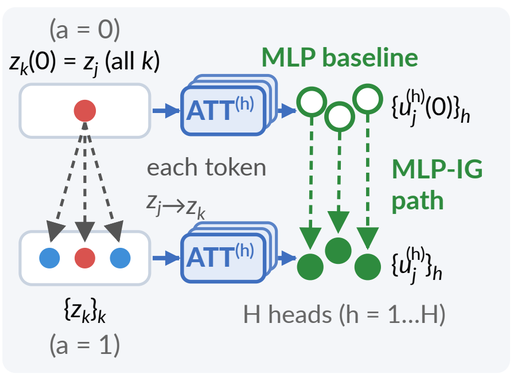
Every path starts at the zero vector $\mathbf{0}$.

Input Token Base (ITB)



All tokens start at the target z_j, so its own $\mathrm{IG}_{j,j} = 0$ (estimated below: zeroRatio / mapRatio).

MLP: ATTITBa=0



At $a = 0$ all tokens are z_j, so each head outputs z_j's value $u_j^{(h)}(0)$: the MLP baseline.

Why ITB? It measures contributions *relative to the target* z_j. Its zero self term [8] is an artifact of the reference, not of the layer (self-attention $\alpha_{j,j} > 0$), so we estimate it (*completion*) and rescale to keep the total: **ITB-zeroRatio** scales the Zero-baseline self term by the other tokens' ITB/Zero ratio; **ITB-mapRatio** splits by $\alpha_{j,j}$:

$$\mathrm{IG}^{\mathrm{zeroRatio}}_{j,j} = \mathrm{IG}^{\mathrm{zero}}_{j,j} \frac{\sum_{k \neq j} \mathrm{IG}^{\mathrm{ITB}}_{k,j}}{\sum_{k \neq j} \mathrm{IG}^{\mathrm{zero}}_{k,j}}, \qquad \mathrm{IG}^{\mathrm{mapRatio}}_{j,j} \approx \left(\sum_{k \neq j} \mathrm{IG}^{\mathrm{ITB}}_{k,j}\right) \frac{\alpha_{j,j}}{1 - \alpha_{j,j}}$$

Why ATTITBa=0? A separate ITB for the MLP would zero its self term again; inheriting the ATT output at the ITB start ($a = 0$) keeps both modules on one reference.

4 Composing ATT and MLP OURS

Chain the two boundaries, LRP-style, and compare the composed token-level allocation with whole-layer IG:

$$\underbrace{\widetilde{\mathrm{IG}}^{\mathrm{LAYER}}_{i,j}}_{\text{whole layer}} \overset{?}{\approx} \underbrace{\sum_h \widetilde{\mathrm{IG}}^{\mathrm{ATT}}_{i,j,h}\, \widetilde{\mathrm{IG}}^{\mathrm{MLP}}_{h,j}}_{\text{composed}}$$

Not an IG chain rule: $J_{\mathrm{LAYER}} = J_u \mathrm{MLP}\, J_{\mathrm{ATT}} + J_z \mathrm{MLP}$ (J_u, J_z: w.r.t. u, z) has a residual route the composition omits; a product of integrals need not equal the integral of a product; and the straight path in u need not be the image of the path in z. We measure this *compositional gap* empirically, not as faithfulness (behavior under perturbation).

5 Result: does the composition match the layer?

Setup. BERT-base-uncased (12 layers, 12 heads); Penn Treebank (PTB) dev (English news), 1,700 sentences from Treebank-3 [9]; IG with $m = 32$ Riemann steps. For each sentence, layer l and output token j, we take the L_2 distance between the two sides of §4:

$$d_j^{(l)} = \left(\sum_i \left(\underbrace{\widetilde{\mathrm{IG}}^{\mathrm{LAYER}}_{i,j}}_{\text{whole layer}} - \underbrace{\textstyle\sum_h \widetilde{\mathrm{IG}}^{\mathrm{ATT}}_{i,j,h}\, \widetilde{\mathrm{IG}}^{\mathrm{MLP}}_{h,j}}_{\text{composed}}\right)^2\right)^{1/2} \quad (\text{lower} = \text{better})$$

Mean over evaluable (sentence, layer, token) triples; top 3 per block; **bold** = best; compare within blocks.

Whole-layer baseline (reference)	Composed: ATT	MLP	L_2 ↓
ITB Input Token Base: all tokens start at z_j	**ITB**	**ATTITBa=0**	**0.305**
	ITB-zeroRatio	ATTITBa=0	0.312
	ITB-mapRatio	ATTITBa=0	0.318
Zero zero vector $\mathbf{0}$	**Zero**	**Zero**	**0.736**
	ITB-zeroRatio	Zero	0.744
	Zero	ATTITBa=0	0.749
LAYER-ITB-zeroRatio whole-layer ITB, self term filled by zeroRatio	**Zero**	**Zero**	**0.735**
	ITB-zeroRatio	Zero	0.742
	Zero	ATTITBa=0	0.748

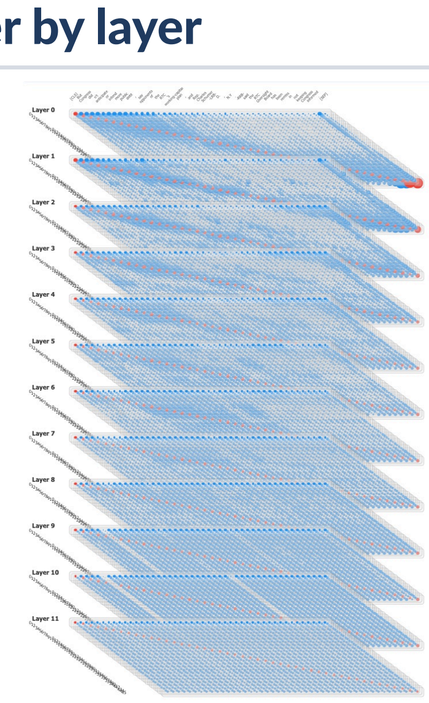
The best MLP baseline follows the reference: ATTITBa=0 under ITB (for every tested ATT baseline); Zero under Zero and LAYER-ITB-zeroRatio.

Over all tested pairs (not only the 3 shown), the module whose baseline moves L_2 more flips: ATT under Zero, MLP under ITB.

6 Use case: reading the flow layer by layer

Whole-layer IG (§3, right cell; Zero baseline), layers 0–11 (0 = embedding output). One PTB dev sentence, qualitative only. Columns (words): layer-input tokens; rows (indices): layer-output tokens; circle size = contribution magnitude; red = from the token itself; blue = from other tokens.

Layer 0: contributions concentrate on each token and its neighbors (phrases); **Layers 9–11:** in this example they appear more evenly spread across the sentence.



Sentence: "But Congress didn't anticipate or intend more public debt, say opponents of the RTC's working-capital plan, and Rep. Charles Schumer (D., N.Y.) said the RTC Oversight Board has been remiss in not keeping Congress informed."

7 Discussion and next steps

In short: LIG measures how well an ATT / MLP split matches whole-layer IG; baseline design matters, and the best MLP baseline follows the reference (ATTITBa=0 or Zero). Needing only module boundaries and gradients, it can be applied to other architectures (untested here).

- **Limits:** BERT-base / PTB only (costly); straight paths may leave the data distribution; ITB completion uniqueness is open; §6 is qualitative; L_2 scalarization discards direction.
- **Next:** decoder-only models, EAP / EAP-IG [10, 11] comparison, task/intervention checks, runtime/memory, baseline guidance, path shape.

[1] Sundararajan+ ICML 2017 · [2] Bach+ PLOS ONE 2015 · [3] Montavon+ LRP overview 2019 · [4] Achtibat+ (AttnLRP) ICML 2024 · [5] Abnar & Zuidema ACL 2020 · [6] Kobayashi+ EMNLP 2020 · [7] Jain & Wallace NAACL 2019 · [8] Sturmfels+ Distill 2020 · [9] Marcus+ Treebank-3, LDC 1999 · [10] Syed+ BlackboxNLP 2024 · [11] Hanna+ COLM 2024. Supported by JSPS KAKENHI JP26K02989, JP26K23861, JP25K24611.